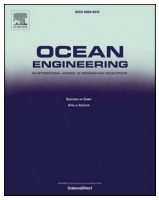
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Research paper



A dynamic risk assessment method for ship conflict based on imperfect information game theory

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ABSTRACT

Ships are unlikely to share their navigation plans at sea, and officers on watch (OOWs) iteratively make decisions to avoid collisions based on limited information and predicted intention. Therefore, ship-ship collision avoidance can be considered a dynamic imperfect information game. Existing conflict risk assessment methods have focused on the uncertainty of ship motion and did not consider game behavior and the bounded rationality of OOWs. This paper proposes a novel dynamic risk assessment method based on the quantal response equilibrium, a modified game-theory model with imperfect information. Expected utility function is employed to quantify the behavior (yielding or not yielding) benefits, incorporating objective risk, energy loss and predicted intention. The OOWs' behavior probability is predicted by considering perception and intention uncertainties. We develop three stages to describe the OOWs' decisions in different risk zones from conflict formation to the end of collision avoidance. Dynamic conflict risk is estimated by calculating the joint probability that OOWs do not yield when encountering ships in the three stages based on the game theory. Simulations and a case study demonstrate that the method realistically reflects conflict risk and reveals ship characteristics influencing yielding. The results provide information for maritime traffic management and OOW training.

1. Introduction

Maritime transportation is crucial in the global trade system, and increasing demand has increased the frequency of accidents. Studies have reported that collision accidents account for nearly 23% of accidents at sea (EMSA, 2022; Li et al., 2021). The consequences of ship collisions are usually catastrophic, posing significant threats to life, property, and the social environment (Liu et al., 2021). Therefore, collision risk analysis has become a hot research topic to ensure maritime safety and reduce the occurrence of collisions.

Many studies have modeled collision risk between ships. Indicator and ship domain (SD)-based methods are mainstream approaches for assessing risk and providing early warning. The former method builds mathematical models to assess collision risk using fuzzy logic (Guan et al., 2020), evidential reasoning (Zhao et al., 2016), support vector machine (Gang et al., 2016), and other techniques. The closest point of approach (CPA), relative velocity, relative bearing, and ship size are parameters in collision risk index (CRI) models. SD models describe ship collision risk levels in a given area around the ship (Xin et al., 2021; Zhang and Meng, 2019). These geometry-based methods provide a comprehensive and reasonable framework for ship collision research. However, most methods assume that the target ship (TS) maintains the same speed and course during the encounter, and the subjective intention of officers on watch (OOWs) to avoid collisions is typically not considered.

Individual ship exhibits dynamic characteristics influenced by the operator's intentions, visibility, the TS's movement, and other external factors. Trajectory prediction method is used to solve these problems and consider the uncertainty of ship movement in risk modeling. The Douglas-Peucker algorithm has been used to identify the trajectory and waypoints to confirm the TS's intention. The probability of conflict has been assessed using nonlinear velocity obstacle (VO) algorithms (Du et al., 2020) and the violation of the SD (Xin et al., 2021). The uncertainty of the TS's action is simplified to the uncertainty of the change in the TS's course. Other methods estimate maneuvering intentions based on real-time observations. The intention probability can be calculated using motion data and a dynamic Bayesian network (Cho and Kim, 2017). Rothmund et al. (2022) inferred the state of multiple intention

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variables based on measured properties to describe different potential trajectories of a ship.

These methods consider changes in ship motion. However, most use an overly optimistic assumption that every ship perceives all circumstances and information similarly. In other words, it is assumed that, OOWs make rational decisions with perfect knowledge of the situation, and errors occur. However, in real scenarios, OOWs consider and evaluate conditions differently. The magnitude of risk perceived by OOWs depends on their risk appetite and the anticipation of the TS's actions, which may be uncertain. They consider the impact of different actions on the outcome and choose the optimal action based on different expected utilities. A collision accident occurred between the oil tanker SANCHI and the bulk carrier CF CRYSTAL in the East China Sea on Jan 6, 2018. According to the accident investigation (MSA, 2018), both OOWs failed to recognize the seriousness of the conflict. Both ships maintained their course because each OOW assumed, their counterpart would yield, which did not occur. Violations of collision regulations (COLREGs) occur frequently, such as the give-way ship maintaining its course and speed, and the stand-on ship changing its course (Wang et al., 2021). Approximately 56% of maritime collisions are associated with uncoordinated collision avoidance due to non-compliance with COLREGs (Statheros et al., 2008).

Ship collision avoidance is associated with significant uncertainty, and collision avoidance decisions and maneuvers reveal game behavior (Wang et al., 2020). Thus, the game theory can be used to analyze ship collision avoidance. Game theory has been widely applied in road traffic analysis to model the decision-making of people involved in vehicle-to-vehicle and pedestrian-to-vehicle interactions (Liu et al., 2023; Wang et al., 2019). Ship encounters typically have longer durations than road traffic encounters, and multiple decisions are required. Therefore, we assume that dynamic game behavior can be used to describe ship encounters. We propose a novel dynamic conflict risk assessment method that considers an OOW's multi-stage decision-making. Based on the potential risk to the ship, we consider three risk zones in the OOW's decision-making stages within different risk acceptance levels. We model non-yielding by the OOW in three stages based on the imperfect information game theory. We introduce the quantal response equilibrium (QRE) to model operator avoidance in three stages based on bounded rationality. Expected utility functions are established to quantify the behavior (yielding or not yielding) benefits to assess the perceived uncertainty of OOWs. The objective collision risk and energy loss are considered. The probability of non-yielding is estimated using a logit quantal response function. The conflict risk from conflict formation to the end of collision avoidance is assessed based on the joint probability function with the assumption that the OOWs on both ships do not yield.

The paper is organized as follows. Section 2 is the literature review. Section 3 describes the proposed model and conflict definition. Section 4 presents the conflict risk assessment methodology based on game models. Section 5 presents case studies to demonstrate the feasibility of the approach. Section 6 provides the results and discussion, and Section 7 concludes the paper.

2. Literature review

2.1. Dynamic conflict risk assessment

Researchers have taken different approaches to model conflict risk and conduct quantitative analysis. Dynamic risk assessment covers different phases of ship operations and is performed in real time (Mehdi et al., 2020). A decision support system is required to deterministically assess risks based on operational parameters, such as speed and relative distance.

2.1.1. Analytical method for risk assessment

The analysis method uses the parameters of the ship's motion during the encounter to calculate the CRI using an analytical equation. Synthetic index methods are popular conflict risk evaluation methods. Mathematical or black-box models have been used for collision risk assessment by incorporating factors describing spatiotemporal relationships near the ship (Liu et al., 2021). Ma et al. (2022) considered the influence of the distance to the CPA (DCPA), the time to the CPA (TCPA), and the relative bearing of the approaching ship. They introduced a collision risk field based on the field theory and established a field strength model using a collision index. Luong et al. (2021) proposed a real-time multi-ship collision risk assessment framework and the maritime traffic hazard index based on the DCPA and TCPA. Abebe et al. (2021) utilized machine learning and Dempster-Shafer evidence theory to establish CRIs based on the DCPA, TCPA, relative distance, relative bearing, and ship speed. The method had high computational efficiency. Goerlandt et al. (2015) presented a theoretical framework with 16 indices to estimate collision risks in different navigation environments. An improved risk index based on virtual collision risk overlay was proposed, considering the ships' geometric parameters (Liu et al., 2023). Yu et al. (2021) proposed a direction-constrained space-time prism-driven approach that considers the current and subsequent ship maneuver characteristics to estimate for multi-ship collision risks. Xin et al. (2023) extended the application of the complex network theory, considering ship motion dynamics. They proposed a multi-scale collision risk estimation model to quantify the criticality of multi-ship collision risk. Methods with multiple indices are commonly used to determine the collision risk probabilities of ships.

Another representative method is SD-based method. The SD refers to the area around a ship that other ships or obstacles should not enter to ensure safe navigation (Xin et al., 2021). Influencing factors include the ships' speed, dimensions, and relative bearing (Lei et al., 2021; Zhang et al., 2021). The shape and size of the SD are determined by empirical estimates (Szlapczynski and Szlapczynska, 2017), fuzzy logic (Zhou and Zheng, 2019), mathematical equations (Zhang and Meng, 2019), and other methods. The collision risk is then assessed based on the intrusion into the SD. Im and Luong (2019) developed a potential risk ship domain by calculating the dynamic potential collision risk (PCR) levels. Szlapczynski and Szlapczynska (2021) considered the degree of domain violation and the relative speed of the two ships in the collision risk assessment.

The VO method has become the most popular collision risk assessment approach. It has several advantages in predicting the potential collision of ships with varying speeds. The VO algorithm assumed nonlinear ship motion and considered ship maneuverability to detect collision hazards (Yuan et al., 2021; Chen et al., 2018, 2020). However, it was challenging to integrate this algorithm into risk assessment models because it increased the computational cost of mapping the spatiotemporal proximity between ships to the velocity space.

This risk assessment models based on objective data provide a reference for evaluating maritime accidents, such as ship collisions, and serve as an early warning for the crews. However, they do not consider subjective risk factors due to human actions, such as voluntariness, perceptual abilities, and risk tolerance, resulting in purely mathematical and objective assessment results. Considering ship motion deviations due to uncertainties is a more realistic approach.

2.1.2. Probabilistic risk assessment considering navigational uncertainty

The dynamics and uncertainties associated with the potential motion of ships are inevitable and influenced by navigation plans, intentions, environmental disturbances, and technological errors. Thus, it is necessary to develop a conflict risk estimation model that incorporates the spatiotemporal uncertainties of ship motion. Trajectory prediction methods are typically employed to address these uncertainties. These methods assume that ships navigate under relatively ideal conditions and predict the future motion of the target ship using various techniques. For instance, historical trajectory analysis has been conducted using probability obstacle methods (Blaich et al., 2015), dynamic

Bayesian networks (Cho and Kim, 2017), neural networks (Liu et al., 2021), and other approaches. From the perspective of ship cooperation, Xin et al. (2021) conducted collision modeling between own ship (OS) and TS. They proposed a probabilistic conflict detection method to estimate the potential collision risk encountered by ships. The method quantifies the uncertainty of trajectories using automatic identification system (AIS) data and relies on information exchange between ships to obtain TS trajectory information. Li et al. (2022) proposed a novel real-time multi-ship conflict risk analysis model. A probabilistic VO model was utilized to incorporate the uncertainty of the TS's motion into the collision risk model. Wang et al. (2023) proposed a ship trajectory estimation model based on deep learning that considered non-cooperative multiple ships in complex water areas. They used the Gaussian distribution to represent uncertainty and evaluated potential collision risks under non-cooperative scenarios based on Monte Carlo simulations. Gao et al. (2023) developed a probability model for long-term prediction based on multivariate Gaussian assumption. They utilized the predicted trajectories for dynamic conflict risk assessment.

These methods consider the uncertainty of TS motion in the risk domain based on Gaussian processes (Rong et al., 2019) and Wiener processes (Wang et al., 2022). The risk level can be accurately predicted by using reliable data and unbiased perception and decision-making. In this study, we propose a probabilistic risk assessment method by developing an incomplete information game theory model to relax the assumptions and assess risk by modeling human behavioral characteristics. The findings can help OOWs and maritime authorities detect conflict risk in complex maritime traffic areas.

2.2. Game theory

The game theory has unique advantages in describing the interactions between two or more players because it captures participant communication. Games can be classified as perfect and imperfect information games based on the extent to which players have knowledge about the other players. At EconPort (Peters, 2016), perfect information is defined as follows: "By perfect information we mean that anything that may impact a buyer or seller's decision-making process is known and understood." As a result, imperfect information refers to incorrect, incomplete or missing information. Different models can be constructed based on different scenarios.

Game theory models are widely used in road traffic analysis to model decision-making in multi-vehicle interactions (Li et al., 2022; Wang et al., 2019) and pedestrian-vehicle interactions (Liu et al., 2023; Sun et al., 2022). Traditional game models assume complete information and perfect rationality of all participants, resulting in Nash equilibrium solutions representing the optimal strategy combinations for participants (e.g., yielding or not yielding). To relax these assumptions, McKelvey and Palfrey (1995) proposed an imperfect information game model called QRE to model interactions among bounded rational individuals. For example, Arbis and Dixit (2019) applied the QRE to model lane-changing maneuvers between an on-ramp driver and a mainline driver.

The use of game theory is less prevalent in maritime shipping. Differential games have been used to control ship safety (Lisowski, 2020; Lisowski and Mohamed-Seghir, 2019) and determine optimal trajectories. Liu et al. (2019) applied cooperative game theory to compute the risk contribution of different ships in multi-ship encounters. They employed game theory for trajectory optimization and global collision risk assessment but did not consider the limitations related to the operators' subjectivity and cognitive abilities during the encounter.

In ship-ship interactions, one player's decision is a response to the other player. Ship encounters typically have longer durations than road traffic encounters, requiring multiple decisions. Therefore, we model the intentions of OOWs using a three-stage decision-making model. In summary, it is feasible to use a game theory model to capture the interactions between OOWs for encountering ships. Since dynamic interference occurs in the encounter between two ships, it is crucial to assess the conflict risk caused by the non-yielding behavior of ships until collision avoidance becomes impossible.

2.3. Motivation and contribution

Several studies developed risk assessment models that considered the uncertainty of ship motion, implicitly considering the rational behavior of OOWs. However, the perceptual ability and risk tolerance of operators significantly influence collision avoidance decisions. Therefore, the game behavior of OOWs in terms of whether to yield or not needs to be considered in collision risk assessment.

The contribution of this paper is twofold. First, this study introduces game theory into conflict risk assessment, considering the bounded rationality of OOWs in situation cognition process. In addition, we propose a three-stage dynamic game model to predict conflict probability. The dynamic risk assessment method combines imperfect information game theory and expected utility theory, and can model the strategy selection by a bounded rational OOW during a ship encounter in three risk zones. The proposed model can help vessel traffic services (VTS) identify potentially dangerous TSs for different ship pairs.

3. Ship collision avoidance considering game behavior

3.1. Overview of meeting ship behavior

When two ships encounter each other at sea, and an early warning occurs, such as violation of the SD (Wang et al., 2023), the two ships interact dynamically to avoid a collision. If they take no avoidance measures until they reach the CPA, a conflict will occur. In serious cases, ship destruction and loss of life may result. Fig. 1 shows an example of conflict identification. Ships A and B are in conflict if ship A maintains its course, and ship B does not yield from encounter moment t_c to moment t_d after entering the danger zone and to the final decision point t_{vd}. The three decision-making stages are the cautious zone (CZ), dangerous zone (DZ) and very dangerous zone (VDZ); they are described in detail in Section 3.2. In addition, we use the minimum distance to collision (MDTC) as the final moment of decision-making by OOWs (Montewka et al., 2010). The value indicates the minimum distance at which the ship is able to perform an effective collision avoidance maneuver. It is closely related to the rate of turn (ROT).

Clearly, ship collision avoidance is a decision process over time and dynamic game behavior rather than a one-time solution. A schematic diagram of the game behavior of ship encounters and its influencing factors is shown in Fig. 2. Two ships constitute the two players in the game. At any given time, too many variables exit in the COLREGs to provide a specific time or distance at the beginning of a given phase and to determine when to act, although the ships are classified as the give-way ship and the stand-on ship. It would be difficult to mutually communicate specific conflict resolution algorithms or intentions in practical applications. The give-way ship may attempt to maintain its course against the rules, and the stand-on ship may be aware of the intention of the give-way ship and respond to it (yielding or not yielding). Thus, each ship has two strategies, i.e., the give-way ship can choose 'to keep' or 'not to keep', and the stand-on ship can choose 'to yield' or 'not to yield'.

Different strategies result in different utilities and different consequences. The strategies are selected based on a comprehensive analysis, bringing about a complex gaming situation. Wickens et al. (2020) observed a balance between safety and efficiency in ship collision avoidance. In addition, it also has to predict the intentions of the TS.

Objective collision risk estimation is the first consideration, i.e., can the other ship pass within a safe distance? This behavior characterizes the utility of the operator, who has chosen to give way. In fact, the operator's judgment of collision risk depends largely on the DCPA and TCPA (Yu et al., 2022). These parameters are explicitly listed in the

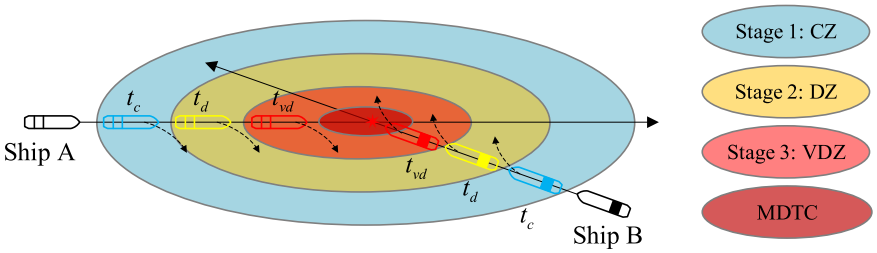


Fig. 1. Definition of ship conflict.

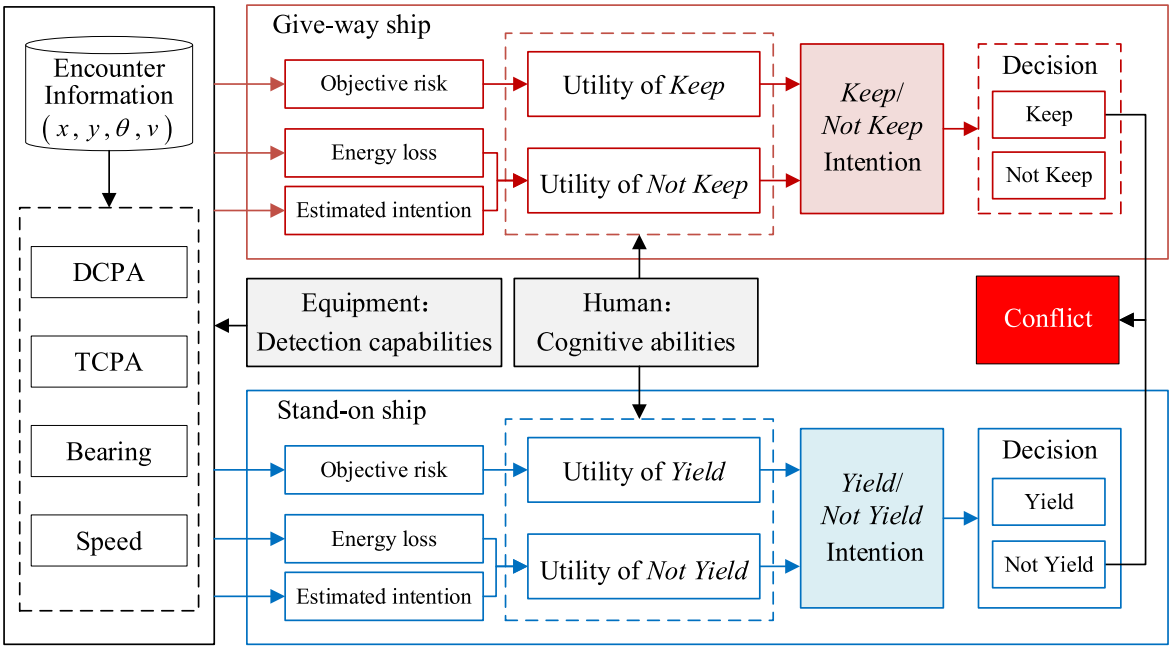


Fig. 2. The game behavior of ship encounters and its influencing factors.

COLREGs. For example, COLREG Rule 7 (b) recommends using radar equipment to identify the DCPA and TCPA as an early warning of collision risk. The DCPA and TCPA represent the spatial and temporal proximity between a pair of ships and can be calculated from the ship's position and velocity vectors.

In addition, a decision to stay on course depends on economic gains and estimated intention of the TS. On the one hand, they do not want to deviate too far from the planned route resulting in fuel consumption. Fuel usage is a major cost component of marine transportation (Zhou et al., 2018). The operator's decision to maintain the course, i.e., the give-way ship maintains its course, and the stand-on ship does not yield, is directly related to fuel consumption. On the other hand, OOW will also estimates the TS's intention to give way and adjust its decision. The OOW of the give-way ship has a belief that the interactive stand-on ship will choose strategy 'to yield', whereas the OOW of the stand-on ship has a belief that the interactive give-way ship will choose strategy 'to keep'.

The utility determines the outcome of the game. The final action can be determined if accurate utilities are estimated for each ship. Internal and external factors contribute to the uncertainty of the utility, substantially affecting the outcome of the game. The OOWs obtain navigation information based on automatic radar plotting aid displays and AIS devices, enabling them to determine encounters. The sensing and detection capabilities depend on the navigation equipment, resulting in information asymmetry between ships. In addition, since the operators make the final decision, their cognitive levels determine the utilities. When the give-way ship makes 'keep' decision, and the stand-on ship makes 'not yield' decision because of the game, a ship conflict occurs.

3.2. Conflict definition and its criticality measure

Ship-ship collision avoidance is essentially a process of weighing its own or collective interests in a conflict, and has distinctive game characteristics. In this paper, conflicts include both near-miss and collision scenarios. We focus on conflict probabilities of failing to yield and breaking the COLREGs because operators make wrong decision due to gaming.

We consider the rules of the COLREGs and make the following assumptions to obtain accurate probabilities.

Assumption 1. The action only refers to the alteration of the course (Li et al., 2021).

According to Rule 8 of the COLREGs, turning alone may be the most effective action to avoid a close-quarters situation under most circumstances.

Assumption 2. The decision-making of OOWs during ship encounters occurs in three zones.

When a risk of collision first begins to apply, there are three stages relating to the permitted or required action for each ship. Many studies have proposed different risk areas for early warning (Du et al., 2020; Im and Luong, 2019), similar to our idea. Therefore, we assume that only the decisions at three zones are considered. OOWs are likely to make less frequent decisions in the short term, although it is possible to have multiple interactions between OOWs during the encounter. According to the potential risk ship domain proposed by (Im and Luong, 2019), who

considered the ship length, speed, and navigation conditions in the SD, we consider three zones with different potential risk levels as boundaries of three decision stages (Fig. 1). In addition, the minimum distance to collision is used as the final boundary to determine the final moment for making a collision avoidance decision (Montewka et al., 2010).

4. Methodology: a game theory-based conflict risk assessment approach

A novel dynamic conflict risk assessment method based on game theory is designed to provide realistic quantitative results to support the crew and maritime safety authorities in understanding the conflict risk of ship motion. The ship's characteristics, traffic conditions, and ship dynamics are considered. The method can determine the operator's intention and quantify the conflict risk from the current moment to when spatiotemporal trajectories intersect. The framework of the proposed approach is shown in Fig. 3. Three decision-making stages are identified first, given the existence of potential risks through the ship domain. The avoidance behaviors of the OOWs are modeled by introducing QRE, which is a modified game-theory model with relaxed assumptions of rational behavior and perfect information in the game environment. Several factors cause deviations in the operator's decision. We construct expected utility functions to characterize the player's perceived uncertainty. Therefore, the decisions of the bounded rational operators are stochastic. Choice probability is estimated by using a logit quantal response function. Finally, the conflict risk is estimated using a joint probability function consisting of the probabilities of making decisions in the three stages dynamic game.

4.1. Quantal response equilibrium model

This section elaborates on the QRE model, which is used to

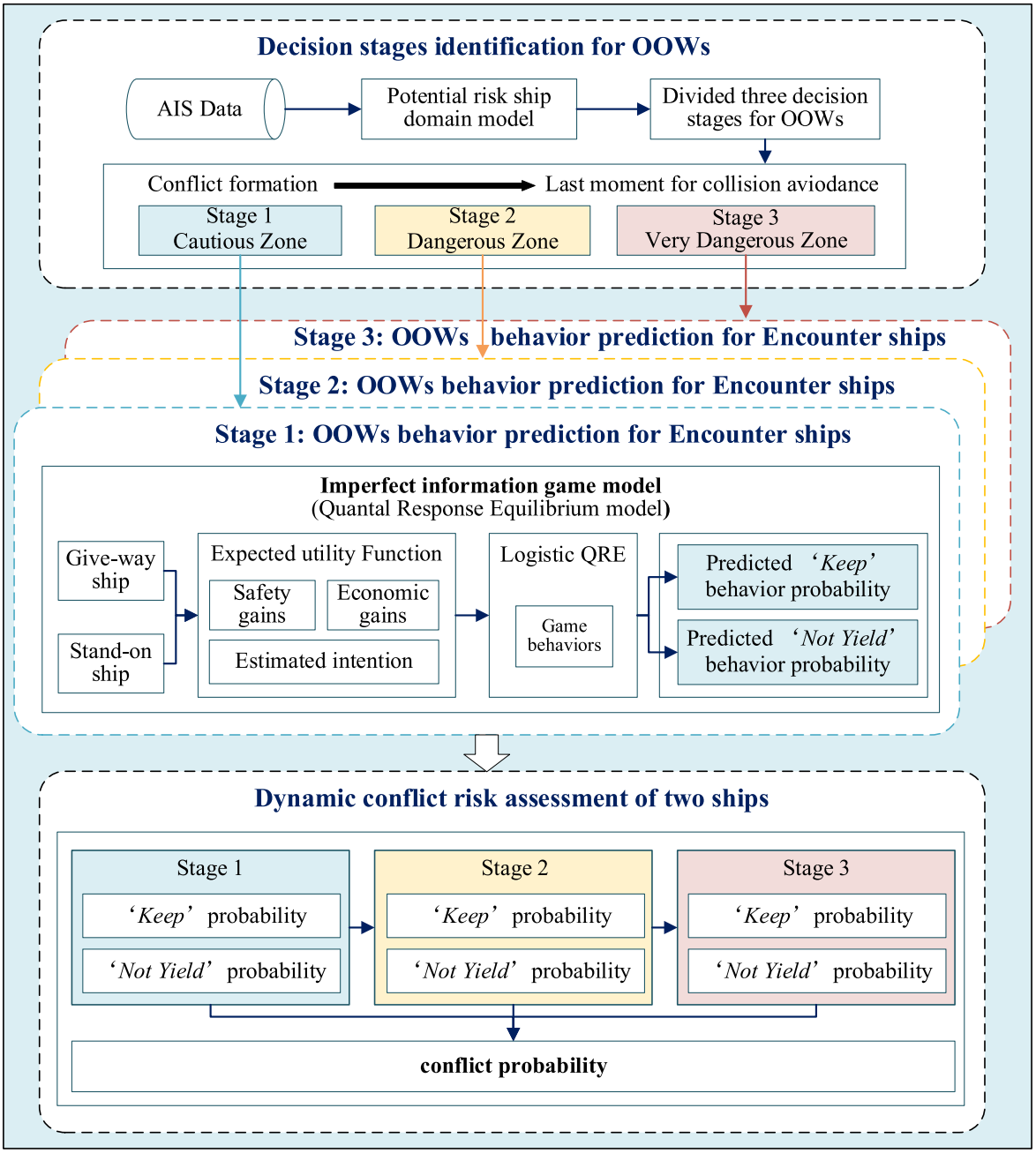


Fig. 3. Framework of the proposed approach.

determine the predicted behavior probability of the ships, considering uncertainty. A normal form of a game and logit quantal response function constitute the model. The model design is detailed in the following sections.

4.1.1. Model formulation

The QRE is based on games with imperfect information, meaning that each player is uncertainty about the actions of another player. This game can be expressed as follows: $\Gamma = \{N, \{S_i\}_{i\in N}, \{P_i\}_{i\in N}, \{u_i(S_i, P_{-i})\}_{i\in N}\}$. $N = \{1, 2, \cdots, n\}$ is a set of players, and for each player i has a strategy set $S_i = \{s_{i1}, s_{i2}, \cdots, s_{iJ}\}$ consisting of J actions. $P_i = \{p_{i1}, p_{i2}, \cdots, p_{iJ}\}$ is the set of probability measures for S_i. Elements of P_i have the form $p_{ij} \geq 0$ and $\sum_j p_{ij} = 1$. These probabilities can be perceived by other players, forming their beliefs about the actions of others. For each i, a payoff function exists: $u_{ij}(p_{-i}) \in u_i(p_{-i})$, which represents the payoff of i choosing the strategy s_{ij}, and all other players choosing their strategy of p. The payoff functions depend on the players' strategies and beliefs; thus, the ship-ship interaction can be quantified.

When two ships enter the risk zone, the give-way ship may attempt to maintain its course against the rules, and the stand-on ship responds yielding or not yielding. We establish the game model using QRE and estimate the probabilities of the strategies of the OOWs using the logit functions. Each player has two possible strategies, i.e., the give-way ship can maintain or not maintain its course, and the stand-on ship can choose to yield or not to yield. Thus, there are two choice probabilities for each player.

1. The set of players: $N = \{\text{give} - \text{way ship} = 1; \text{stand} - \text{on ship} = 2\}$.
2. The set of strategies for each player $S_i = \{s_{ij}\}$: $S_1 = \{s_{11} = \textit{Keep}, s_{12} = \textit{Not Keep}\}$, $S_2 = \{s_{21} = \textit{Yield}, s_{22} = \textit{Not Yield}\}$.
3. The set of choice probabilities for each player $P_i = \{p_{ij}\}$: $P_1 = \{p_{11} = p_{keep}, p_{12} = 1 - p_{keep}\}$, $P_2 = \{p_{21} = p_{yield}, p_{22} = 1 - p_{yield}\}$ which can be perceived by other players.
4. The payoff function for each player $u_i = \{u_{ij}(p_{-i})\}$: $u_1 = \{u_{11}(p_{-1}), u_{12}(p_{-1})\}$, $u_2 = \{u_{21}(p_{-2}), u_{22}(p_{-2})\}$ (p_{-1} denotes stand-on ship adopts "yield" strategy of p_{yield} and p_{-2} denotes give-way ship adopts "keep" strategy of p_{keep}).

The payoff functions are related to the player's strategies and beliefs. To characterize the above decisions, the payoff function is modeled based on the expected utility theory, with which the utility of a player depends on the anticipation of another player's strategy. The choice probability can be estimated using the logit quantal response functions.

4.1.2. Logistic quantal response equilibrium

In the QRE, the utility of each player for each strategy is subject to error. Based on the principle of maximum utility, player i chooses action j if the utility is maximized. We define such an error set, known as the set of choice behaviors in a risky environment, as follows:

$$\sigma_{ij}(u_i) = \{\varepsilon_i | u_{ij} + \varepsilon_{ij} \geq u_{ik} + \varepsilon_{ik}, \forall k \in 1, \cdots, J\} \tag{1}$$

where u_{ij} and u_{ik} are the utilities of player i for strategy j and k; ε_{ij} and ε_{ik} are the utility errors of player i for strategy j and k.

There are two implications for interpreting the error terms.

- Taste variation for differences between ship operators in the weights of the explained attributes.
- Operation' estimation errors, they cannot accurately assess the actual risk and efficiency.

Given probability P of the players' belief which represents all other players choosing their strategy of p , each set $\sigma_{ij}(u_i(p))$ identifies the errors region that will cause player i to choose strategy s_{ij}. For a QRE, the probability of player i choosing strategy s_{ij} given u_i can be expressed by the following equation, which is called the quantal response function:

$$p_{ij} = \int_{\sigma_{ij}(u_i)} f(\varepsilon) d\varepsilon \tag{2}$$

where $f(\varepsilon)$ is the error distribution.

The logistic quantal response function has been widely used in estimating the choices of individuals. It assumes that the player's anticipation has errors that follow the extreme value distribution (McKelvey and Palfrey, 1995). The probability of the player's strategy is estimated by the logit function.

$$p_{ij} = \frac{e^{u_{ij}(p)}}{\sum_{k=1}^{J} e^{u_{ik}(p)}} \tag{3}$$

4.2. Expected utility function

The objective of the expected utility function is to quantify the strategy benefits of players and facilitate the prediction of behavior probabilities during ship encounters. The expected utility theory can be used to analyze situations where individuals make decisions without knowing the resulting outcomes, i.e., decision-making under uncertainty. These individuals will choose the action with the highest expected utility, which is the weighted average of all utilities for all possible outcomes under the given circumstances. This method can be used to quantify the strategy benefits of players. The normal form of the expected utility function is as follows:

$$EU = \sum_{i=1}^{N} p_n U(x_n) \tag{4}$$

where $U(x_n)$ is the utility of outcome x_n, and p_n is the probability of outcome x_n.

In ship-ship interactions, the ship operator is motivated to minimize the risk of conflict and reduce energy loss. Therefore, there are two outcome utilities for each strategy, i.e., the economic and safety benefit and the player's utility depend on the anticipation of another player's strategy. For example, when the give-way ship does not maintain its course or the stand-on ship yields, they are motivated safety benefits. When a give-way ship maintains its course or the stand-on ship does not yield, they are motivated by economic gains, but it is assumed that the TS will yield. That is, it relies on a belief in the actions of the TS. The expected utility functions for the actions during ship encounters are expressed in Eqs. (5)–(8).

Give-way ship's utilities:

$$EU_{Keep} = p_{yield} \times U^1_{economic} \tag{5}$$

$$EU_{DoNotKeep} = U^1_{safety} \tag{6}$$

Stand-on ship's utilities:

$$EU_{Yield} = U^2_{safety} \tag{7}$$

$$EU_{DoNotYield} = \left(1 - p_{keep}\right) \times U^2_{economic} \tag{8}$$

where EU_{Keep} and $EU_{DoNotKeep}$ are the give-way ship's expected utilities of maintaining and not maintaining its course, EU_{Yield} and $EU_{DoNotYield}$ are the stand-on ship's expected utilities of yielding and not yielding. $U^1_{economic}$ is the economic benefit of the give-way ship, U^1_{safety} is the safety benefit of the give-way ship, $U^2_{economic}$ is the economic benefit of the stand-on ship, and U^2_{safety} is the safety benefit of the stand-on ship.

The critical factor that significantly affect the actions must be identified and modeled to quantify the benefit of ship avoidance strategies. Studies have investigated the influence of various factors, such as perceived collision risk, ship speed, ship length, ship gross tonnage, and the navigation environment (Li et al., 2021; Park et al., 2020; Ugurlu and Cicek, 2022; Weng et al., 2018). Therefore, the proposed utilities should have a multivariate nature to account for the spatiotemporal interactions during ship encounters. In this study, the utilities of the meeting ships are expressed as follows:

Give-way ship's utilities:

$$EU_{Keep} = p_{yield} \times a_1 * v_{gw}^2 \tag{9}$$

$$EU_{DoNotKeep} = \lambda_1 - \left(a_2 * DCPA_{gw} + a_3 * TCPA_{gw}\right) \tag{10}$$

Stand-on ship's utilities:

$$EU_{Yield} = \lambda_2 - \left(b_1 * DCPA_{so} + b_2 * TCPA_{so}\right) \tag{11}$$

$$EU_{DoNotYield} = \left(1 - p_{keep}\right) \times b_3 * v_{so}^2 \tag{12}$$

where v_{gw} is the give-way ship's speed, and v_{so} is the stand-on ship's speed, a_1 and b_3 are coefficients of the kinetic energy variable v^2, which are linearly proportional to the gross tonnage of the ship, $DCPA_{gw}$ and $TCPA_{gw}$ are the risk metrics related to the navigation equipment on the give-way ship, $DCPA_{so}$ and $TCPA_{so}$ are the risk metrics related to the navigation equipment on the stand-on ship, p_{keep} and p_{yield} are the perceived probabilities of the operator's decision. a_2, a_3, b_1, and b_2 are coefficients of the risk metrics, λ_1 denotes the safety margins that the OOWs of the give-way ship can accept and λ_2 denotes the safety margins that the OOWs of the stand-on ship can accept, which describe the boundaries of the risk on the latent variable (DCPA and TCPA) scale.

Based on the kinetic energy theorem, the economic benefit is characterized as the kinetic energy variable v^2. It is related to the degree of fuel loss. The gross tonnage of the ship indicates the ship's mass and normalize it as a coefficient with the range [0,1]. In other words, we assume that the coefficients a_1 and b_3 in the utility function are linearly proportional to the gross tonnage of the ship. We assume that 75,000 can be used to carry out the normalization for ships higher than 75,000 gross tonnage or more have the same impact on navigator's perceived risk (Chin and Debnath, 2009). The efficiency gain is also related to the steering angle, but since we do not consider specific steering operations, this gain is ignored.

The safety benefit is inseparable from the conflict risk and is closely related to the DCPA and TCPA. The higher the collision risk, the greater the safety benefit of giving way. The safety benefit is obtained by subtracting the operator's safety threshold from a linear combination of variables. The values were obtained from the literature (Debnath and Chin, 2010) and are listed Table 1 and Table 2. The ships are clustered into several categories according to their gross tonnage to consider the effects of the ship size and navigational conditions on risk perception. The risk is modeled separately for days and nights.

4.3. Decision process and dynamic conflict risk estimation

In this section, we describe the human decision process for once interaction and dynamic conflict probability assessment methods. Data flow in dynamic risk assessment process is shown in Fig. 4. They are discussed in detail in the following sections.

Table 1
The category for ship with Gross Tonnage (Debnath and Chin, 2010).

Ship category	Description
VC 1	*if* $300 \le GT \le 12000$
VC 2	*if* $12000 < GT \le 20000$
VC 3	*if* $20000 < GT \le 75000$
VC 4	*if* $GT > 75000$

4.3.1. Decision process

The interaction between give-way ship and stand-on ship is modeled as a bounded rationality two-player game. The expected utility of OOWs in each interaction is calculated based on the encounter information. Then, the choice probability is computed for once decision process.

According to the QRE, the choice probabilities of the players are equal to another player's perceived probabilities in Eq. (13) and Eq. (14) (Watling, 2006). The utility function is influenced by the operator's perceptual bias, which is subject to a certain distribution. The logistic quantal response function assumes that the player's anticipation has errors with an extreme value distribution (McKelvey and Palfrey, 1995). Based on the expected utility function, the probabilities of the operator's strategy can be estimated by the logit function.

$$p_{keep} = \frac{\exp\left[EU_{Keep}\left(p_{yield}\right)\right]}{\exp\left[EU_{Keep}\left(p_{yield}\right)\right] + \exp\left[EU_{DoNotKeep}\right]} \tag{13}$$

$$p_{yield} = \frac{\exp\left[EU_{yield}\right]}{\exp\left[EU_{yield}\right] + \exp\left[EU_{DoNotYield}\left(p_{keep}\right)\right]} \tag{14}$$

McKelvey and Palfrey (1995) verified the existence and uniqueness of logistic QRE. It is a fixed-point problem to calculate p_{keep} and p_{yield} mathematically.

4.3.2. Dynamic conflict probability estimation

The first step of the risk assessment is to identify the decision stages. The size of the zone affects the probability of the OOW's decision in that zone, and affects the calculation of conflict probabilities, as shown in Fig. 5. The decision process simulation based on QRE is performed for each stage. Next, the decision probabilities of OOWs are calculated for each decision stage, and the conflict probability can be finally obtained based on the behavioral estimation of the two ships.

(1) Identification of decision stages

The traditional approach assumes that the operator acts only when a certain risk threshold is reached; however, since individual risk acceptance varies, we assume that the operator's intention differs in different risk zones. Im and Luong (2019) proposed a potential risk SD consisting of four areas around the ship with different potential risk levels. The PCR was used to reflect the collision risk of OS for a point in the SD. The higher the PCR, the more dangerous it is. Depending on various PCR, the distance D_φ from OS to surrounding points at different angles, i.e., the extent of the decision stage, is calculated by the following equation. The PCR values for the three stages are listed in Table 3.

$$D_\varphi = \begin{cases} \sqrt{\dfrac{-2 \times \ln(PCR)}{\lambda v^2 \sin\varphi_p^2 + \cos\varphi_p^2}} \times v \times \sigma \times L \ \textit{if}\ 0^\circ \le \varphi_p \le 90^\circ \ \textit{or}\ 270^\circ \le \varphi_p \le 360^\circ \\ \sqrt{\dfrac{-2 \times \ln(PCR)}{\lambda}} \times \sigma \times L \ \textit{if}\ 90^\circ < \varphi_p < 270^\circ \end{cases} \tag{15}$$

where φ_p is the angle between OS and the surrounding point; v is the ship's speed; L is the ship's length; (σ, λ) are the longitudinal and lateral influence parameters that determine the longitudinal and lateral influence ranges of the SD. Their recommended values are (0.35,0.03) in restricted areas and (1.25,0.02) in open areas.

When a ship enters very dangerous area, it also has the possibility of emergency collision avoidance. We determine the last decision moment with the minimum distance to collision (Montewka et al., 2010), indicating that collisions cannot be avoided even if the operator takes action.

Table 2
The coefficients for ship collision risk (Debnath and Chin, 2010).

	VC 1		VC 2		VC 3		VC 4	
	Day	Night	Day	Night	Day	Night	Day	Night
DCPA (Cables length)	0.2660	0.2179	0.5611	0.6502	0.2641	0.2710	0.2431	0.2088
TCPA (minutes)	0.1168	0.0902	0.3278	0.2637	0.1151	0.1181	0.1013	0.0892
λ	3.1519	3.0112	6.9348	8.5806	3.4408	3.9437	3.3680	3.2375

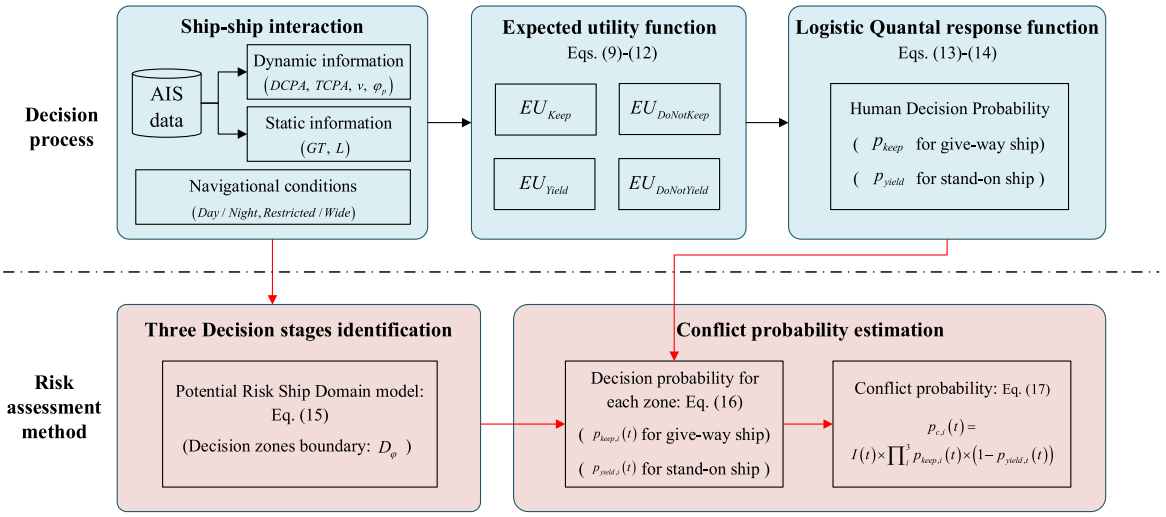


Fig. 4. Data flow in dynamic risk assessment process.

(2) Dynamic conflict risk estimation

During the dynamic risk assessment, that the final risk values are related to the time. Different decisions are made by operators during ship collision avoidance due to different perceived risks of conflict. The highly coupled relationship between the ships involved in the collision avoidance scenario is characterized by three stages logical decision-making structure over time. Each stage corresponds to a different level of PCR. The dynamic conflict risk is estimated using numerical simulation.

Safety and efficiency concerns during ship encounters result in uncertainty about the operators' intentions. Rational players make decisions based on the anticipated actions of other players. After identifying the risk areas, we can calculate the decision probability for each zone by averaging the values with a step size Δt in the simulations.

According to the definition in Section 3.2, a criticality measure is proposed. Ships A and B are regarded in conflict if Eq. (16) is satisfied.

Table 3
Potential collision risk values for three boundaries (Im and Luong, 2019).

Boundaries	PCR
Cautious	0.01
Dangerous	0.4
Very dangerous	0.8

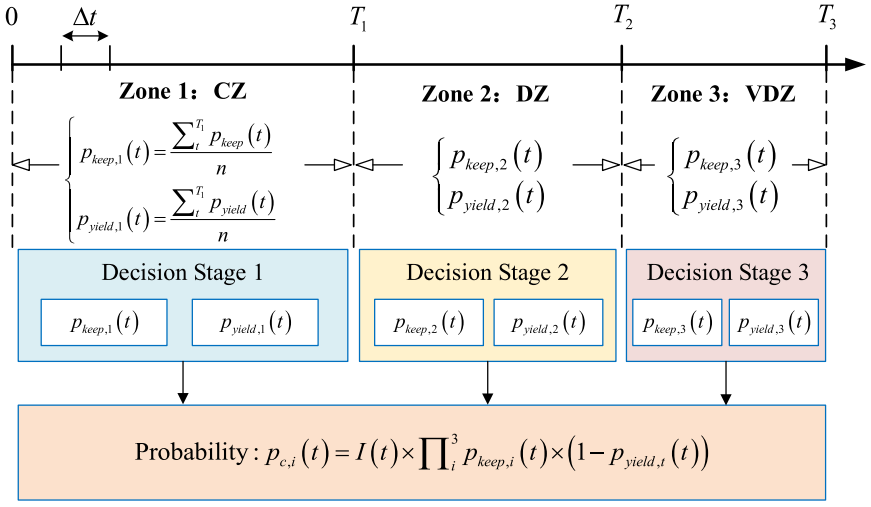


Fig. 5. The relationship between the inclusion of the three zones and the findings.

$$I(t)=\begin{cases}1, A(t\geq t_c)=keep\wedge B(t\geq t_c)=not\ yield\\ 0, else\end{cases} \tag{16}$$

where $I(t)$ denotes an indicator function, and 1 denotes a conflict, $A(t\geq t_c)$ and $B(t\geq t_c)$ are the actions of the ships from encounter moment t_c to the final decision point, at which ship A maintains its course, and ship B does not yield.

The conflict risk at time t is defined as the probability that the two ships do not give way until the last moment. We combine the choice probabilities into a joint probability function to estimate the likelihood of a conflict, as shown in Eq. (17).

$$p_{c,i}(t)=I(t)\times\prod_i^3 p_{keep,i}(t)\times\left(1-p_{yield,i}(t)\right) \tag{17}$$

where $p_{keep,i}(t)$ is the probability that the give-way ship maintains its course in stage i at moment t, and $p_{yield,i}(t)$ is the probability that the stand-on ship alters its course with yields in stage i at moment t.

5. Case studies

Simulations and true case were conducted to assess the conflict risk during a two-ship encounter and validate the feasibility and effectiveness of the proposed method. We simulated a two-ship encounter in Section 5.1 and conducted a case study of a collision accident in a crossing encounter based on historical AIS data in Section 5.2.

5.1. Encounter scenarios based on simulations

We construct two typical encounter situations for near-miss case and collision case in the case of crossing encounter. The initial state sets included positions, speeds, course angles and lengths of the involved ships are listed in Table 4. Fig. 6 illustrates the spatial layout of the two scenarios. For collision case of OS and TS_1, the initial value of the DCPA is 0 NM, and they will collide after 1081 s at TCPA = 1081 s. For near-miss case of OS and TS_2, the initial value of the DCPA is -0.13 NM, and they will reach the minimum distance after 1046 s at TCPA = 1046 s. According to the COLREGs, OS is the give-way ship and TS is the stand-on ship. The algorithm was implemented in MATLAB. The boundaries for the three zones can be measured as the distance between the two ships and calculated according to Equation (15) in Section 4.3.2. With a step size of 10s in the simulations, then we can find the time corresponding to each boundary, as shown in Table 6.

5.2. Accident case based on AIS data

At about 1950LT (1150 UTC) on January 6, 2018, the Panama registered oil tanker SANCHI collided with the Hong Kong (China) registered bulk carrier CF CRYSTAL in East China Sea at approximate position 30°51′.1N/124°57′.6E. SANCHI, loaded with a cargo of condensate oil, was on her voyage from Assaluyeh, Iran to Daesan, Republic of Korea (ROK). CF CRYSTAL was loaded with sorghum in bulk, bounding from Kalama, USA to Dongguan, China. This accident happened in open waters with good visibility, the northeast wind Beaufort force was 4–5, and the sea state was slight. The collision breached the cargo tanks of SANCHI, resulting in the leakage of condensate oil and consequent fire and explosions and eventual sinking of the ship. It was a very serious marine accident, as defined in the Code of the International Standards and Recommended Practices for a Safety Investigation into a Marine Casualty or Incident.

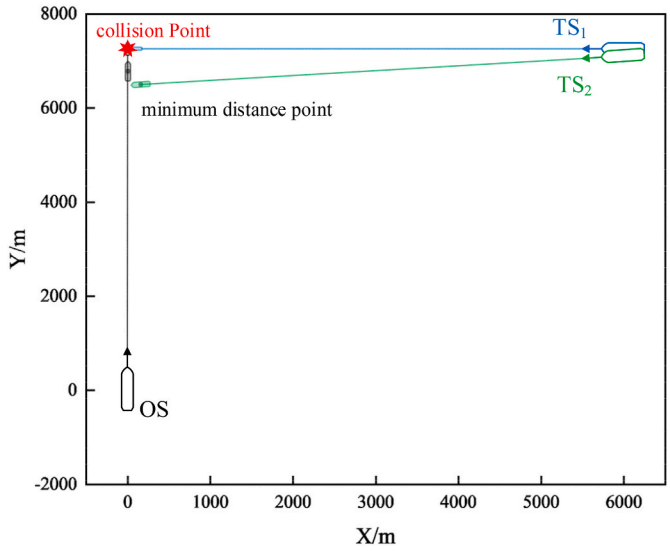


Fig. 6. Spatial layout of scenarios.

According to the COLREGS, the two ships were in a crossing situation. The SANCHI was the give-way ship, and CF CRYSTAL was the stand-on ship. According to the accident investigation, both ships failed to take action, even though they were aware of each other.

Table 5 list basic information on the two ships. At a full load, the SANCHI traveled at 10.8 knots, and the CF CRYSTAL traveled at 13.4 knots. Fig. 7 displays the trajectories of the two ships based on the AIS data (MSA, 2018). The dashed ellipse box marks the collision location. It can be seen that any ship has not acted. We can find the time corresponding to each boundary with the accident statistics, as shown in Table 6.

6. Results and discussion

6.1. Results of simulations

6.1.1. Conflict probability of simulation cases

The conflict probabilities for the near-miss and collision cases are shown in Fig. 8. No matter which cases are validated, the trend of conflict probability is the same. These results indicate that the conflict probability due to the expected unwillingness of OOWs to give way during the period from the time the ship enters the CZ to when the collision between the two ships cannot be avoided. As the relative distance between the two ships decreases, the probability of ship conflict increases, and the rate of increase in conflict probability becomes greater over time, reflecting reality. This finding implies that the OOWs should take action as early as possible, which aligns with the COLREGs.

By comparing the collision case and near-miss case, we notice that the conflict probability is greater in the near-miss case. In general, the probability of conflict should be higher for the near-miss scenario. Because ship conflicts include near-miss and collision scenarios, and near-miss scenarios are more common. Moreover, the near-miss scenarios involving extreme danger but no actual collision, which has a relatively small objective risk, and there is a greater likelihood that the

Table 4
The initial set of encounter situation.

Ship	Position (X, Y) (NM)	Speed (knots)	Course	Length (m)	GT
OS	(0,0)	12.6	000	225	24000
TS_1	(3.2,3.8)	10.7	270	180	12000
TS_2	(3.2,3.8)	10.7	266	180	12000

Table 5
Basic information about the two ships (MSA, 2018).

Ship Detail	SANCHI	CF CRYSTAL
Ship Type	Oil Tanker	Bulk Carrier
Gross Tonnage	85462	41073
Length Overall (m)	274.18	225

Table 6
The distance and time corresponding to boundaries of three zones.

Scenario	Metric	Boundary of each zone		
		CZ	DZ	VDZ
Collision case	Distance (nm)	4.21	1.89	0.93
	Time (s)	170	670	880
Near-miss case	Distance (nm)	4.20	1.83	0.97
	Time (s)	170	660	860
Accident case	Distance (nm)	5.69	2.41	1.19
	Time (s)	1480	2011	2220

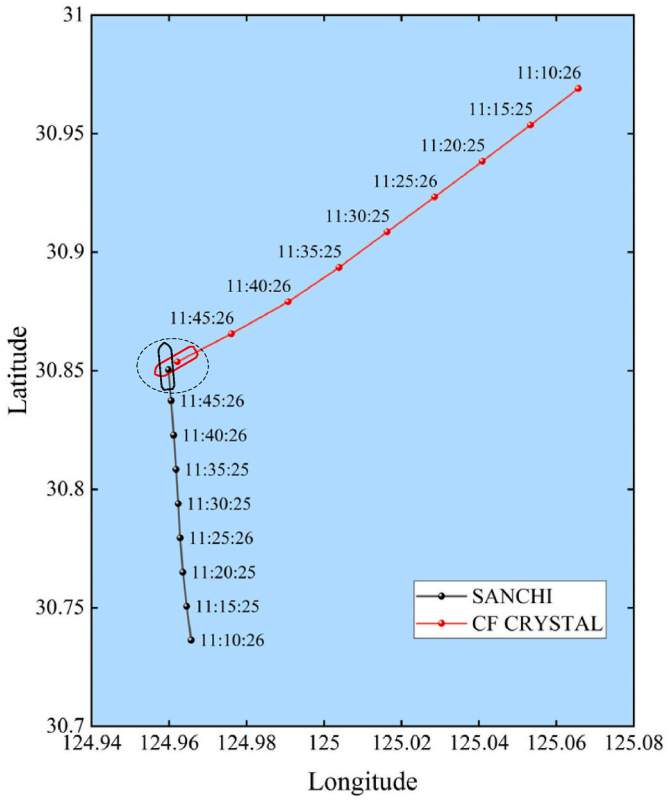


Fig. 7. The trajectories of the two ships.

two ships will not yield, resulting in a higher probability of conflict.

The results indicate that the proposed our method is more sensitive to near-miss scenarios. A lot of research has been done on identifying near-miss scenarios (Liu et al., 2023; Szlapczynski and Szlapczynska, 2021), and our study can further identify the magnitude of conflict probability due to the uncertainty of the perception and intention of OOWs in near-miss scenarios at the same risk level.

6.1.2. Comparison with non-game scenario

In this part, we compare the conflict probability of the collision case in two conditions.

(1) Game scenario: the intention estimation for target ship is considered;
(2) Non-game scenario: the game behavior of the OOW is ignored in the risk assessment, i.e., it is assumed that the operator does not consider the intention of the target ship when making avoidance decision.

The simulation of collision case and accident case are set for the experiment, and the settings of ships are listed in Tables 4 and 5. The conflict probabilities for the two scenarios are shown in Fig. 9. The

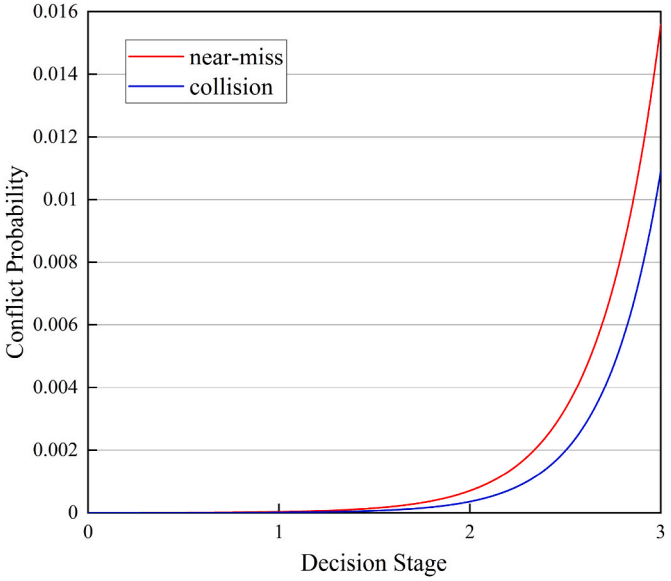


Fig. 8. Conflict probability of simulation cases.

conflict probability is overestimated when the game behavior is ignored. The results demonstrate that operators are more cautious when the game behavior between OOWs is considered. The reason is that individuals only pursue their own interests and do not consider the perspectives of others in a non-game scenario. This finding aligns with other studies, suggesting that operators do not always behave selfishly and in an antisocial manner. Instead, they often forgo short-term gains to cooperate with others (Barmpounakis et al., 2016; Wang et al., 2020).

The differences between considering and not considering game theory are tied to the encounter situations. Under certain scenarios, the differences between the two conditions may not be apparent. The non-game condition is assumed that the operator does not consider the intention of the target ship when making avoidance decision. The decision of OOWs is related only to safety and economic gains. The economic gain has a significant impact on the differences between considering and not considering game theory, specifically ship tonnage or speed affects the probability of conflict under both conditions. When ship tonnage or speed is relatively small, the economic gain is low. Operators act in their own self-interest and are more likely to yield in scenarios where game theory is not considered, resulting in insignificant differences. If the ship's tonnage or speed becomes larger, the difference becomes pronounced.

6.1.3. Comparison with the rule-aware time-varying conflict risk measure and CRI method

We compare the proposed method with the rule-aware time-varying conflict risk (R-TCR) measure and the classic CRI method. The R-TCR evaluates the entire conflict risk incorporating COLREGs, varying maneuverability, and different ship domains (Li et al., 2021). The R-TCR is defined as a ratio of the achievable actions that lead to conflict, see Eq. (18). The traditional CRI remains the fundamental approach to risk assessment in marine navigation. It combines DCPA and TCPA by the weighted method (Kearon, 1977), see Eq. (19). The parameters of the CRI method are set as follows: $\alpha_1 = 1$ and $\alpha_2 = 2$. A crossing situation is used for the experiment. The details for the ships are listed in Table 7.

$$\mathrm{R-TCR} = \frac{n(DA)}{n(AA)} \tag{18}$$

where $n(DA)$ is the number of achievable actions; $n(AA)$ is the number of dangerous actions.

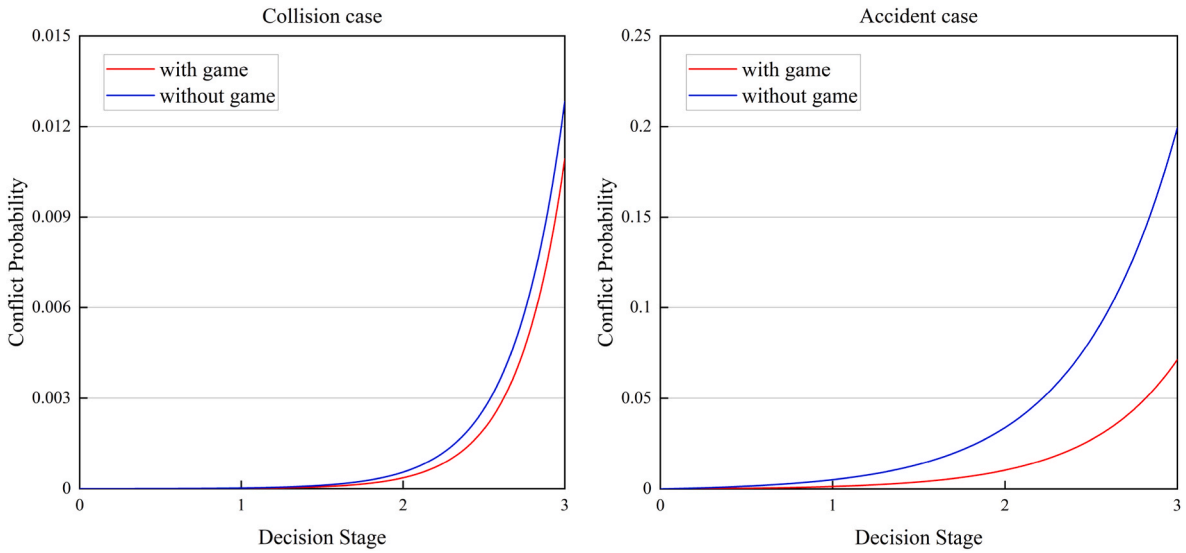


Fig. 9. The difference between game and without game.

Table 7
The setting of experiment.

Ship	Position (X, Y) (NM)	Speed (knots)	Course	Length (m)	GT	DCPA (NM)	TCPA (s)
OS	(0,0)	15.6	000	225	24000	–	–
TS_1	(3,4)	11.7	270	180	12000	0	926

$$CRI^{(t)} = \alpha_1 \times \left(\frac{Max\left(TCPA^{(t)}\right) - TCPA^{(t)}}{Max\left(TCPA^{(t)}\right) - Min\left(TCPA^{(t)}\right)} \right)^2 + \alpha_2 \times \left(DCPA^{(t)}\right)^2 \quad (19)$$

where $DCPA^{(t)}$ is the distance to the closest point of approach at time t; $TCPA^{(t)}$ is the time to the closest point of approach at time t.

The conflict probability value is normalized for comparison. The results in the simulation are shown in Fig. 10. Regardless of which methods are used, their trends over time are the same. However, the two methods have two differences. Firstly, compared with the R-TCR method, the starting point of our method is 0. For reason that we focus mainly on the conflict resulting from decision process among OOWs, i.e., we are concern about conflict probabilities of failing to yield from conflict formation to the end of collision avoidance because operators make wrong decision due to gaming. There is a chance to make up for a poor decision at the beginning. The R-TCR is concerned with the probability of collision due to dangerous evasive actions after making the correct decision. The starting moment means that the probability that collision can not be avoided by only changing the OS's course is not 0. This method provides a reference for our future research, considering both the decisions of OOWs and the actions taken after those decisions. Secondly, the three methods have different physical significance. The CRI method combines DCPA and TCPA, which are non-dimensional parameters. Thus, the CRI has no physical significance. The R-TCR represent the collision probability that the ship can not avoid the dangers, which indicates the difficulty to avoid collisions (Li et al., 2021). In contrast, our method can represent the probability of conflict due to the dynamic game of OOWs for encountering ships, which indicates the severity of OOWs decision-making error. In summary, the proposed method is suitable for analyzing the risk distribution and behaviors of OOWs.

6.2. Results of accident case

As shown in Fig. 11, both ships remain inactive during the encounter. As the relative distance between the two ships decreases, the conflict probability increases, and the rate of increase in the conflict probability becomes greater over time, which is in line with reality.

The following results are observed when comparing the game based on imperfect and perfect information (Fig. 12).

(1) The perfect information game model assumes that the officer has complete and accurate information. It determines the optimal strategic. The imperfect information game model considers the effect of information incompleteness and provides a probability

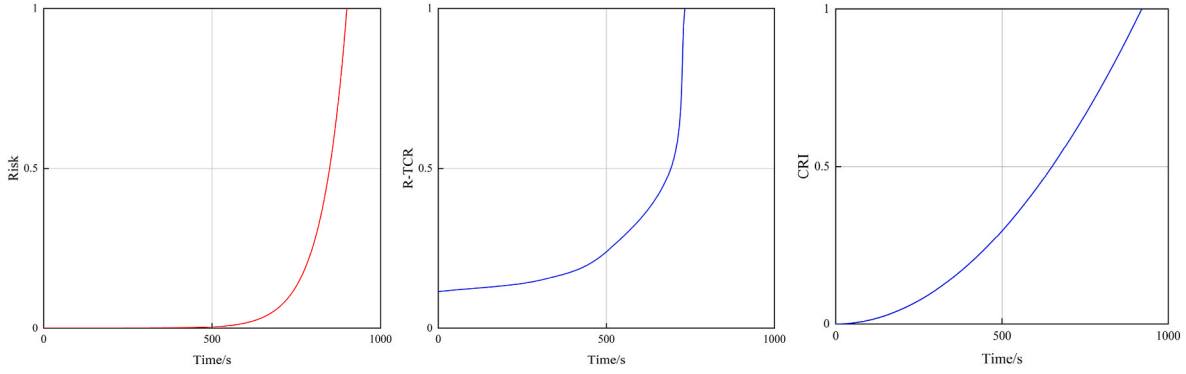


Fig. 10. The difference with the R-TCR and CRI method.

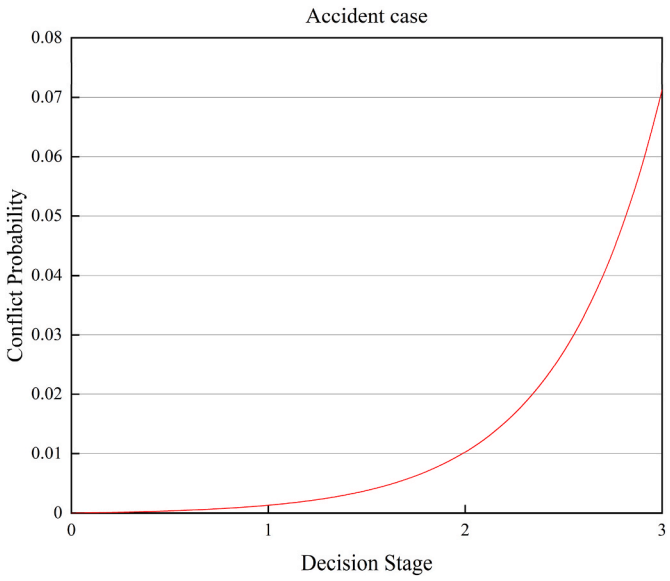


Fig. 11. Conflict probability of the accident case.

distribution for the officer's strategy. This indicates that in a real situation, the officer may make different strategic choices according to different situations when facing the conflict risk, rather than an absolutely certain choice. In this case, the SANCHI tended to yield behavior ($P > 0.5$) later than the CF CRYSTAL because its tonnage was twice that of the CF CRYSTAL, and the difference in speed was non-significant. The SANCHI considered economy more. According to the accident report, the SANCHI did have a lower level of attention.

(2) There is a difference between the probability distribution of strategy choice in the imperfect information game model and the optimal strategy choice in the perfect information game model. The reason is that the uncertainty associated with imperfect information results in randomness in the strategic choice of the officer. In contrast, the model based on perfect information provides the deterministic optimal strategy.

The results indicate that the imperfect information game model better reflects uncertainty and randomness in real situations. It considers the effect of information incompleteness and provides the probability distribution of the officer's strategy. This method enables more comprehensive risk assessment and decision analysis to help officers cope with conflict risks. Therefore, the proposed model better reflects real conditions and has higher applicability and practicality.

6.3. *Influences of gross tonnage and speed on operator behaviors*

Since yielding by one ship does not directly cause collisions to occur, we discuss the influences of the gross tonnage and speed on the single-ship behavior in the game. The simulation of collision case is set for the experiment. The ship parameters are listed in Table 4. Different values are used for the gross tonnage and speed of the TS to observe the effect on ship interactions.

Fig. 13 shows the effect of the TS tonnage on ship's probability of not yielding. As the TS tonnage increases, the probability of not yielding gets larger at each decision stage because it indicates a stronger willingness to not turn for efficiency. On the contrary, the probability of not yielding for OS becomes smaller and smaller, because the OOW becomes more cautious as the tonnage of TS increases.

The effect of the different speeds of the TS on each ship's behavior of not yielding are showed in Fig. 14. As the TS speed increases, the probability of not yielding for OS and TS is increasing at each decision stage. First, the TS's willingness to not yield increases due to increasing costs. Second, as the speed increases, the DCPA of both ships rises, the objective collision risk decreases, and the probability of not yielding increases for both ships.

Further, we discuss the effects of changes in tonnage and speed on the probability of conflict. Fig. 15 illustrates the effect of the TS tonnages on the conflict probability. As the tonnage increases, the conflict probability increases, because the ships tend to maintain their course to optimize their travel efficiency.

Fig. 16 shows the effect of the ships' speeds on conflict probability. As the ship speed increases, the distance between the two ships decreases faster. This results in the ships entering the risk zone earlier, leading to a higher conflict probability. This is because as the speed increases, ships are more inclined to maintain their course, and higher speeds make it easier to change the heading.

In addition, compared to Fig. 15, the effect of change in speed on conflict probability is greater than the effect of tonnage, for reasons that can be analyzed in terms of behavioral change. A increase in TS tonnage causes a decrease in the OS's probability of not yielding in the same decision stage, whereas a increase in TS speed increases the probability. The ship tonnage and speed significantly influence the operators' yield intentions, particularly for high-tonnage and high-speed ships, consistent with other studies (Wickens et al., 2020).

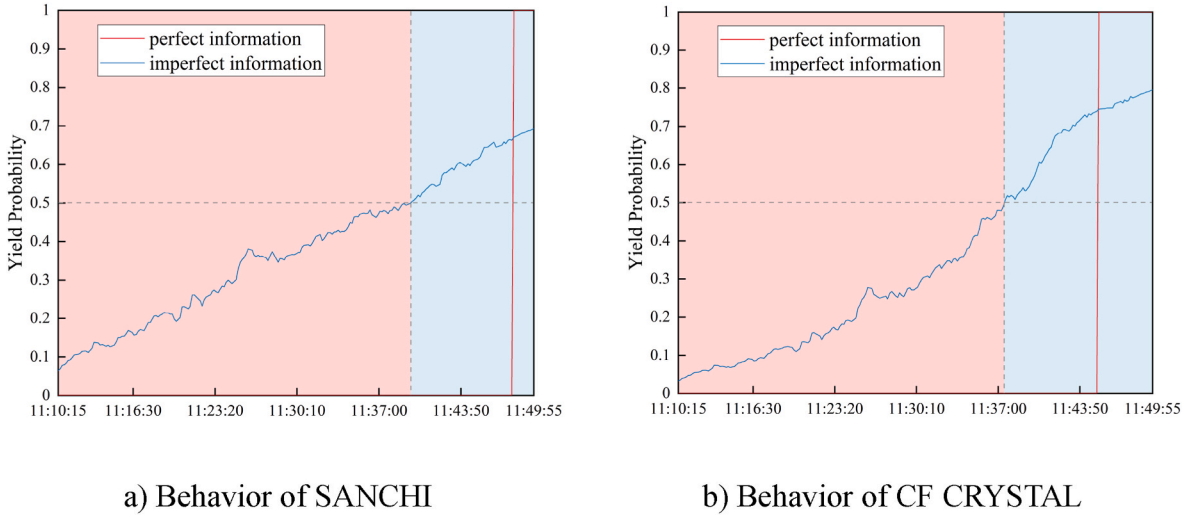


Fig. 12. The yield probability of the two ships.

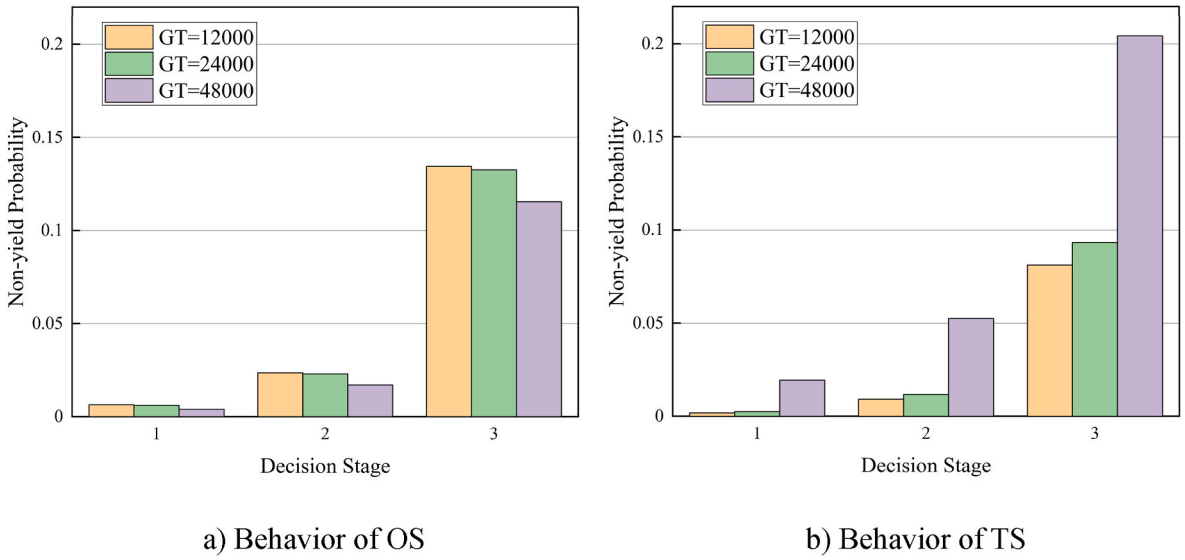


Fig. 13. The probability of not yielding behavior of ships under different TS tonnage.

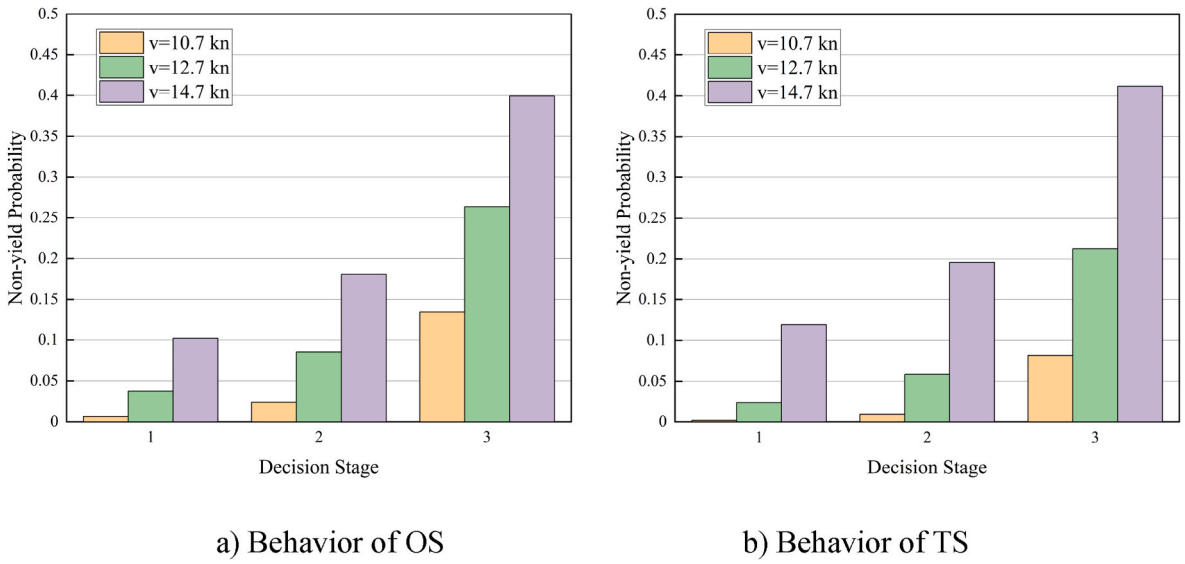


Fig. 14. The probability of not yielding behavior of ships under different TS speeds.

6.4. *Sensitivity analysis*

Based on conflict probability estimation Equation (17), the relationship between the explanatory variables and the probability of conflict is indirect. Thus, a sensitivity analysis was conducted to determine the effects of the variables. The values of the variables were changed by 10%, 20% and 50%, and the other variables remained unchanged. The simulation of collision case is set for the experiment, and the settings of ships are listed in Table 4. The following four steps were used.

1. For the process of ship encounter, the original conflict probability P_c for the three decision stages are calculated based on the steps in Section 4.3, given the observed values of DCPA, TCPA, v_{gw}, and v_{so}.
2. Changing the values of the given predictor variables by 10%, 20% and 50% to calculate the new conflict probabilities $P_c^{'}$ using Equation (17).
3. Calculating changes in conflict probability $\left(P_c^{'} - P_c\right)/P_c$ for the three decision stages.
4. Repeating steps 1–3 for the other explanatory variables and plotting charts, as shown in Fig. 17.

Based on the results of sensitivity analysis, we can draw some conclusions from two aspects.

(1) Factor ranking in the same decision stage

By observing the variation within each stage of Fig. 17, the importance ranking of the influencing factors for a 10%, 20% and 50% increases are the same, which are shown in Table 8. In the case of collision in crossing situations, the sensitivity of the speed of the give-way ship is greater than that of the stand-on ship. This is because the speed of the give-way ship in the simulation case is greater than the speed of the stand-on ship. High speed ships have higher kinetic energy in encounter situations. In addition, the effect of the speed is greater than that of the CPA, and the effect of TCPA is greater than that of DCPA. Thus, the time factor significantly affects the game behavior of the OOWs.

The higher sensitivity of the TCPA compared to the DCPA suggests

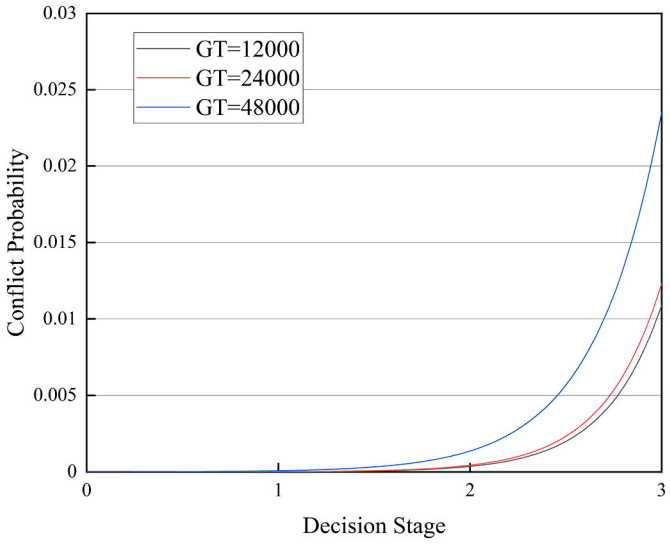


Fig. 15. Conflict probability for different TS tonnages in collision scenario.

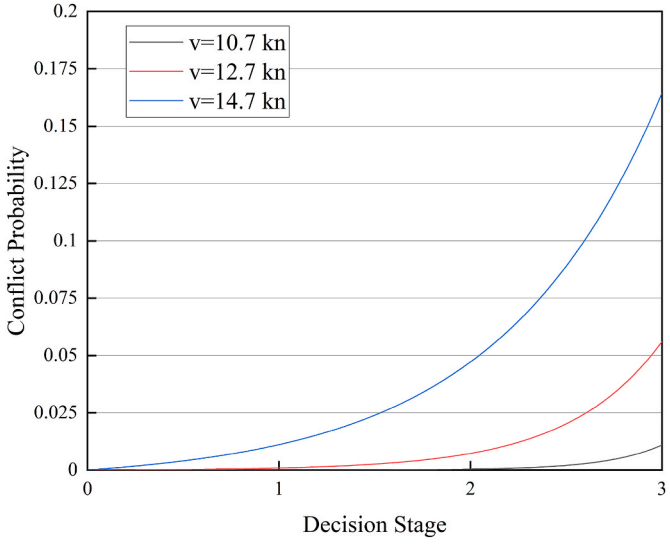


Fig. 16. Conflict probability for different speeds in collision scenario.

that changes in the time of ship encounters have a more significant effect on collision risk assessment than changes in the distance between ships. This can be attributed to the critical role of time in determining the potential for collision avoidance maneuvers. Changes in the time of ship encounters have a greater impact on navigational decisions and risk perception. When TCPA increases, OOWs feel that there is still sufficient time to react and the likelihood of not giving way increases, thus increasing the likelihood of conflict. This coincides with the fact that the timing of collision avoidance is an important influence in maritime navigation. The danger can certainly be discussed only if the encounter distance reaches the DCPA threshold, and our findings aim to emphasize the temporal dimension of ship encounters and its crucial role in collision avoidance decisions.

(2) Factor sensitivity variation in different decision stages

From each line in Fig. 17, it can be noticed that for the same risk factor, the more dangerous the decision area is, the less sensitive a factor is. This is because the higher the objective risk, the more likely it is that the operator follows the rules for avoidance, which relatively does not cause much delay in the game for the encountering ships. The larger the relative distance, the more complex the psychology of the game.

In summary, these results can be used to refine the game theory model to adjust parameters. The results indicate that the speed and TCPA should be closely monitored to mitigate the conflict risk and enhance situational awareness of stakeholder during maritime navigation, as they can greatly influence yielding behaviors of OOWs.

6.5. *Potential limitations and further improvements*

In traditional risk analysis, they often described conflict risk in terms of urgency in time and distance (Liu et al., 2023; Ma et al., 2022), so as to provide early warning for the crew. This study proposed a novel framework to assess the conflict risk caused by subjective intentions of the crew. We focus more on the probability that not yielding leads to a conflict. The applicability conditions of our model are: (1) If the two ships do not give way they will enter the conflict zone, which will lead to conflict including near-miss and collision scenarios; (2) During the collision avoidance both ships do not intend to give way due to gaming and breaking the rules. After complying with the rules and choosing to give way at any point in the three decision zones we have divided, the behavior probability of not giving way is 0. Further conflict risk studies are needed to explore the behavioral characteristics of OOWs in the cases involving operations that conform to the COLREGs.

In addition, our study does not directly mention traditional statistical techniques. Our main contribution is the study of collision risk assessment methods based on game theory. Statistical analysis can be carried out based on the statistical data. It is difficult to apply existing data

Table 8
The importance ranking (1–4) of various influencing factors in the assessment process.

	DCPA	TCPA	v_{gw}	v_{so}
Stage 1 (CZ)	4	2	1	3
Stage 2 (DZ)	4	3	1	2
Stage 3 (VDZ)	4	3	1	2

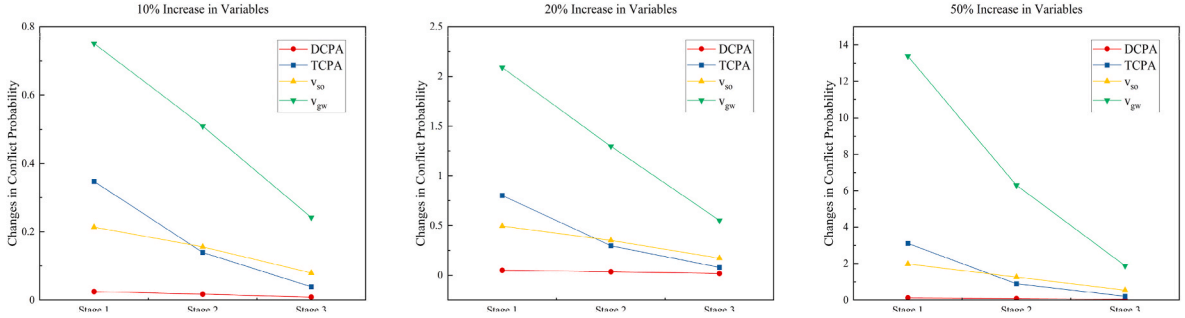


Fig. 17. Sensitivity analysis results for different change in the risk variable.

because this method needs to identify the yield decision points, which cannot be obtain directly through AIS data. Further data mining is required, while may have large errors. Therefore, it is best to obtain the data through experimental statistics. When we get the yield decision moment and its corresponding encounter information, we can use maximum likelihood estimation to estimate the risk parameters of the utility in the model. Specifically, using utility differences as potential estimators, it can be solved iteratively by the Expectation Maximization algorithm with the goal of maximizing the sum of the log-likelihood functions of the two ships. Further research can also consider other risk factors that are closely related to navigational safety, e.g., ship types, sizes, maneuverability and ROT, etc.

Moreover, there are also some limitations in this study. Firstly, the perceived risk capacity and decision-making of OOWs are affected by different encounter scenarios and the uncertainty of environmental factors, such as varying meteorological conditions (winds, waves, currents, visibility), the complexity of a water area (restricted, crowded). Our model does not differentiate between them, which may be further analyzed through scenario clustering. Secondly, individual characteristics such as operator experience and fatigue levels may influence personnel expected utility, warranting attention in future research for considering these factors to enhance the accuracy and reliability of conflict assessment. Finally, encounters involving multiple ships are common in high-density waters, ship may have different roles (Liu et al., 2023; Yu et al., 2022), making it difficult to determine which ship will act first. Modeling and analyzing conflict risk assessment considering multi-ships games necessitates further research to strengthen maritime safety management.

7. Conclusion

The continuous interaction between encountering ships creates a complex situation, especially the uncertainty of the OOW's intentions, posing significant challenges in analyzing conflict risk. In this paper, we developed a conflict risk assessment method based on the dynamic game theory that considers the bounded rationality of individual. The proposed method has several characteristics: (1) modeling personnel behavior and analyzing behavioral characteristics, (2) considering uncertainties such as perception and decision-making biases, and (3) evaluating conflict risks until the officer realizes that avoidance is difficult. The ships' actions and the OOWs' intentions were investigated and quantified in different stages, enabling maritime risk analysis.

The effectiveness of the proposed method was validated through simulations of crossing scenarios and real case study. The results demonstrated that the proposed model accurately reflected conflict risks during ship encounters in crossing scenarios. Furthermore, the method accurately estimates the OOWs' intentions during ship encounters. The proposed game theory approach is practical for conflict risk assessment, assisting VTS in identifying high-risk ship encounters.

In the future, we aim to conduct a more comprehensive analysis, mining additional ship behavior characteristics from AIS data, to enhance the feasibility of the proposed method. In addition, our model is difficult to model the game behavior in the multi-ship scenarios, further studies are required to enrich its applicability in various complex multi-ship environments.

CRediT authorship contribution statement

Jie Zhao: Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Conceptualization. **Shengkui Zeng:** Writing – review & editing, Supervision, Conceptualization. **Jianbin Guo:** Writing – review & editing, Validation, Supervision. **Haiyang Che:** Writing – review & editing, Supervision, Methodology, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

Abebe, M., Noh, Y., Seo, C., Kim, D., Lee, I., 2021. Developing a ship collision risk index estimation model based on Dempster-shafer theory. Appl. Ocean Res. 113, 102735.
Arbis, D., Dixit, V.V., 2019. Game theoretic model for lane changing: incorporating conflict risks. Accid. Anal. Prev. 125, 158–164.
Barmpounakis, E.N., Vlahogianni, E.I., Golias, J.C., 2016. Modeling cooperation and powered-two wheelers short-term strategic decisions during overtaking in urban arterials. International Journal of Transportation Science and Technology 5 (4), 227–238.
Blaich, M., Köhler, S., Reuter, J., Hahn, A., 2015. Probabilistic collision avoidance for vessels. IFAC-PapersOnLine 48 (16), 69–74.
Chen, P., Huang, Y., Mou, J., van Gelder, P.H.A.J., 2018. Ship collision candidate detection method: a velocity obstacle approach. Ocean. Eng. 170, 186–198.
Chen, P., Huang, Y., Papadimitriou, E., Mou, J., van Gelder, P.H.A.J., 2020. An improved time discretized non-linear velocity obstacle method for multi-ship encounter detection. Ocean. Eng. 196, 106718.
Chin, H.C., Debnath, A.K., 2009. Modeling perceived collision risk in port water navigation. Saf. Sci. 47 (10), 1410–1416.
Cho, Y., Kim, J., 2017. Collision Probability Assessment between Surface Ships Considering Maneuver Intentions. OCEANS 2017. Aberdeen.
Debnath, A.K., Chin, H.C., 2010. Navigational traffic conflict technique: a proactive approach to quantitative measurement of collision risks in port waters. J. Navig. 63 (1), 137–152.
Du, L., Goerlandt, F., Valdez Banda, O.A., Huang, Y., Wen, Y., Kujala, P., 2020. Improving stand-on ship's situational awareness by estimating the intention of the give-way ship. Ocean. Eng. 201, 107110.
EMSA, 2022. Annual Overview of Marine Casualties and Incidents 2022. European Maritime Safety Agency.
Gang, L., Wang, Y., Sun, Y., Zhou, L., Zhang, M., 2016. Estimation of vessel collision risk index based on support vector machine. Adv. Mech. Eng. 8 (11), 2071836213.
Gao, D., Zhu, Y., Guedes Soares, C., 2023. Uncertainty modelling and dynamic risk assessment for long-sequence ais trajectory based on multivariate Gaussian process. Reliab. Eng. Syst. Saf. 230, 108963.
Goerlandt, F., Montewka, J., Kuzmin, V., Kujala, P., 2015. A risk-informed ship collision alert system: framework and application. Saf. Sci. 77, 182–204.
Guan, Z., Wang, Y., Zhou, Z., Wang, H., 2020. Research on early warning of ship danger based on composition fuzzy inference. Jmse 8 (12), 1002.
Im, N., Luong, T.N., 2019. Potential risk ship domain as a danger criterion for real-time ship collision risk evaluation. Ocean. Eng. 194, 106610.
Kearon, J., 1977. Computer Program for Collision Avoidance and Track Keeping. Mathematical Aspects of Marine Traffic. Academic Press INC. LTD, London, UK.
Lei, J.Y., Liu, L., Chu, X.M., He, W., Liu, X.L., Liu, C., 2021. Automatic identification system data-driven model for analysis of ship domain near bridge-waters. J. Navig. 74 (6), 1284–1304.
Li, G., Weng, J., Hou, Z., 2021. Impact analysis of external factors on human errors using the arbn method based on small-sample ship collision records. Ocean. Eng. 236, 109533.
Li, M., Mou, J., Chen, L., He, Y., Huang, Y., 2021. A rule-aware time-varying conflict risk measure for mass considering maritime practice. Reliab. Eng. Syst. Saf. 215, 107816.
Li, M., Mou, J., Chen, P., Rong, H., Chen, L., van Gelder, P.H.A.J., 2022. Towards real-time ship collision risk analysis: an improved R-tcr model considering target ship motion uncertainty. Reliab. Eng. Syst. Saf. 226, 108650.
Li, N., Yao, Y., Kolmanovsky, I., Atkins, E., Girard, A.R., 2022. Game-theoretic modeling of multi-vehicle interactions at uncontrolled intersections. IEEE Trans. Intell. Transport. Syst. 23 (2), 1428–1442.
Lisowski, J., 2020. Game control methods comparison when avoiding collisions with multiple objects using radar remote sensing. Rem. Sens. 12 (10), 1573.
Lisowski, J., Mohamed-Seghir, M., 2019. Comparison of computational intelligence methods based on fuzzy sets and game theory in the synthesis of safe ship control based on information from a radar arpa system. Rem. Sens. 11 (1).
Liu, K., Wu, X., Zhou, Y., Yuan, Z., Yang, X., Xin, X., Zhuang, S., 2023. A conflict cluster-based method for collision avoidance decision-making in multi-ship encounter situations. Ocean. Eng. 288, 116038.
Liu, K., Yuan, Z., Xin, X., Zhang, J., Wang, W., 2021. Conflict detection method based on dynamic ship domain model for visualization of collision risk hot-spots. Ocean. Eng. 242, 110143.

Liu, M., Kolmanovsky, I., Tseng, H.E., Huang, S., Filev, D., Girard, A., 2023. Potential game-based decision-making for autonomous driving. IEEE Trans. Intell. Transport. Syst. 24 (8), 8014–8027.
Liu, Z., Wu, Z., Zheng, Z., 2019. A cooperative game approach for assessing the collision risk in multi-vessel encountering. Ocean. Eng. 187, 106175.
Liu, Z., Wu, Z., Zheng, Z., Yu, X., Bu, X., Zhang, W., 2023. A domain-based model for identifying regional collision risk and depicting its geographical distribution. J. Mar. Sci. Eng. 11 (11), 2092.
Liu, Z., Zhang, B., Zhang, M., Wang, H., Fu, X., 2023. A quantitative method for the analysis of ship collision risk using ais data. Ocean. Eng. 272, 113906.
Luong, T.N., Hwang, S., Im, N., 2021. Harbour traffic hazard map for real-time assessing waterway risk using marine traffic hazard index. Ocean. Eng. 239, 109884.
Ma, W., Wang, H., Wang, S., 2022. Critical collision risk index based on the field theory. J. Mar. Sci. Eng. 10 (11).
McKelvey, R.D., Palfrey, T.R., 1995. Quantal response equilibria for normal form games. Game. Econ. Behav. 10 (1), 6–38.
Mehdi, R.A., Baldauf, M., Deeb, H., 2020. A dynamic risk assessment method to address safety of navigation concerns around offshore renewable energy installations. Proc. IME M J. Eng. Marit. Environ. 234 (1), 231–244.
Montewka, J., Hinz, T., Kujala, P., Matusiak, J., 2010. Probability modelling of vessel collisions. Reliab. Eng. Syst. Saf. 95 (5), 573–589.
MSA. Report on the investigation of the collision between M.T. Sanchi and M.V. Cf crystal in East China sea. Maritime Safety Administration of P.R.China (MSA). Retrieved from. https://www.mardep.gov.hk/en/msnote/pdf/msin181 7anx1.pdf.
Park, D., Yim, J., Yang, H., Lee, C., 2020. Navigators'errors in a ship collision via simulation experiment in South Korea. Symmetry-Culture and Science 12 (4), 529.
Peters, G., 2016. In: Flores, V., Gomide, F., Janusz, A., Meneses, C., Miao, D., Peters, G., Ślęzak, D., Wang, G., Weber, R., Yao, Y., Flores, V., Gomide, F., Janusz, A., Meneses, C., Miao, D., Peters, G., Ślęzak, D., Wang, G., Weber, R., Yao, Y. (Eds.), A Rough Perspective on Information in Extensive Form Games. Springer International Publishing, Cham, pp. 145–154.
Rong, H., Teixeira, A.P., Guedes Soares, C., 2019. Ship trajectory uncertainty prediction based on a Gaussian process model. Ocean. Eng. 182, 499–511.
Rothmund, S.V., Tengesdal, T., Brekke, E.F., Johansen, T.A., 2022. Intention modeling and inference for autonomous collision avoidance at sea. Ocean. Eng. 266, 113080.
Statheros, T., Howells, G., McDonald-Maier, K., 2008. Autonomous ship collision avoidance navigation concepts, technologies and techniques. J. Navig. 61 (1), 129–142.
Sun, X., Lin, K., Wang, Y., Ma, S., Lu, H., 2022. A study on pedestrian-vehicle conflict at unsignalized crosswalks based on game theory. Sustainability 14 (13).
Szlapczynski, R., Szlapczynska, J., 2017. Review of ship safety domains: models and applications. Ocean. Eng. 145, 277–289.
Szlapczynski, R., Szlapczynska, J., 2021. A ship domain-based model of collision risk for near-miss detection and collision alert systems. Reliab. Eng. Syst. Saf. 214, 107766.
Ugurlu, H., Cicek, I., 2022. Analysis and assessment of ship collision accidents using fault tree and multiple correspondence analysis. Ocean. Eng. 245, 110514.
Wang, M., Wang, Y., Cui, E., Fu, X., 2023. A novel multi-ship collision probability estimation method considering data-driven quantification of trajectory uncertainty. Ocean. Eng. 272, 113825.
Wang, S., Zhang, Y., Huo, R., Mao, W., 2022. A real-time ship collision risk perception model derived from domain-based approach parameters. Ocean. Eng. 265, 112554.
Wang, S., Zhang, Y., Zheng, Y., 2021. Multi-ship encounter situation adaptive understanding by individual navigation intention inference. Ocean. Eng. 237, 109612.
Wang, T., Wu, Q., Zhang, J., Wu, B., Wang, Y., 2020. Autonomous decision-making scheme for multi-ship collision avoidance with iterative observation and inference. Ocean. Eng. 197, 106873.
Wang, X., Liu, Y., Wang, F., Liu, Z., Zhao, H., Xin, J., 2019. Driver's lane selection model based on multi-player dynamic game. Adv. Mech. Eng. 11 (1), 2072051078.
Watling, D., 2006. User equilibrium traffic network assignment with stochastic travel times and late arrival penalty. Eur. J. Oper. Res. 175 (3), 1539–1556.
Weng, J., Li, G., Chai, T., Yang, D., 2018. Evaluation of two-ship collision severity using ordered probit approaches. J. Navig. 71 (4), 822–836.
Wickens, C.D., Williams, A., Clegg, B.A., Smith, C.A.P., 2020. Nautical collision avoidance. Hum. Factors 62 (8), 1304–1321.
Xin, X., Liu, K., Loughney, S., Wang, J., Li, H., Ekere, N., Yang, Z., 2023. Multi-scale collision risk estimation for maritime traffic in complex port waters. Reliab. Eng. Syst. Saf. 240, 109554.
Xin, X., Liu, K., Yang, Z., Zhang, J., Wu, X., 2021. A probabilistic risk approach for the collision detection of multi-ships under spatiotemporal movement uncertainty. Reliab. Eng. Syst. Saf. 215, 107772.
Yu, H., Fang, Z., Murray, A.T., Peng, G., 2021. A direction-constrained space-time prism-based approach for quantifying possible multi-ship collision risks. IEEE Trans. Intell. Transport. Syst. 22 (1), 131–141.
Yu, H., Meng, Q., Fang, Z., Liu, J., Xu, L., 2022. A review of ship collision risk assessment, hotspot detection and path planning for maritime traffic control in restricted waters. J. Navig. 75 (6), 1337–1363.
Yu, Q., Teixeira, A.P., Liu, K., Guedes Soares, C., 2022. Framework and application of multi-criteria ship collision risk assessment. Ocean. Eng. 250, 111006.
Yuan, X., Zhang, D., Zhang, J., Zhang, M., Guedes Soares, C., 2021. A novel real-time collision risk awareness method based on velocity obstacle considering uncertainties in ship dynamics. Ocean. Eng. 220, 108436.
Zhang, F., Peng, X., Huang, L., Zhu, M., Wen, Y., Zheng, H., 2021. A spatiotemporal statistical method of ship domain in the inland waters driven by trajectory data. Jmse 9 (4), 410.
Zhang, L., Meng, Q., 2019. Probabilistic ship domain with applications to ship collision risk assessment. Ocean. Eng. 186, 106130.
Zhao, Y., Li, W., Shi, P., 2016. A real-time collision avoidance learning system for unmanned surface vessels. Neurocomputing 182, 255–266.
Zhou, D., Zheng, Z., 2019. Dynamic fuzzy ship domain considering the factors of own ship and other ships. J. Navig. 72 (2), 467–482.
Zhou, K., Chen, J., Liu, X., 2018. Optimal collision-avoidance manoeuvres to minimise bunker consumption under the two-ship crossing situation. J. Navig. 71 (1), 151–168.